

Statistical Modelling and Inference

Theory test

Spring 2023

You have a total of 80 minutes to complete this test. Do not begin this test until you are ready to start.

- This is an open book test, however you MUST NOT collaborate with other students or ask for help on the Internet. By completing this test, you agree to the academic honesty policy.

1. Let $Y \sim \text{Bin}(n, p)$ and consider two estimators for p ; namely

$$\hat{p}_1 = \frac{Y}{n} \text{ and } \hat{p}_2 = \frac{Y+1}{n+2}.$$

a. Show that $\hat{p}_1$ is unbiased for p .

[3 marks]

b. Derive the bias of $\hat{p}_2$.

[3 marks]

c. Find $MSE_{\hat{p}_1}(p)$ and $MSE_{\hat{p}_2}(p)$.

[8 marks]

d. If $p = 0.5$, which estimator is preferred by MSE? Justify your answer.

[2 marks]

[Question total: 16]

Solutions:

i.

$$\begin{aligned} E[\hat{p}_1] &= E[Y/n] \\ &= \frac{1}{n} E[Y] \\ &= \frac{np}{n} \\ &= p. \end{aligned}$$

ii.

$$\begin{aligned} b_{\hat{p}_2}(p) &= E[\hat{p}_2] - p \\ &= E\left[\frac{Y+1}{n+2}\right] - p \\ &= \frac{np+1}{n+2} - p \\ &= \frac{np+1-p(n+2)}{n+2} \\ &= \frac{1-2p}{n+2}. \end{aligned}$$

iii.

$$\begin{aligned}
 MSE_{\hat{p}_1}(p) &= \text{Var}(\hat{p}_1) + b_{\hat{p}_1}(p)^2 \\
 &= \text{Var}(Y/n) + 0 \\
 &= \frac{1}{n^2} \text{Var}(Y) \\
 &= \frac{np(1-p)}{n^2} \\
 &= \frac{p(1-p)}{n}.
 \end{aligned}$$

$$\begin{aligned}
 MSE_{\hat{p}_2}(p) &= \text{Var}(\hat{p}_2) + b_{\hat{p}_2}(p)^2 \\
 &= \text{Var}\left(\frac{Y+1}{n+2}\right) + \left(\frac{1-2p}{n+2}\right)^2 \\
 &= \frac{\text{Var}(Y) + (1-2p)^2}{(n+2)^2} \\
 &= \frac{np(1-p) + (1-2p)^2}{(n+2)^2}.
 \end{aligned}$$

iv.

$$\begin{aligned}
 MSE_{\hat{p}_1}(p) &= \frac{1}{4n} \\
 MSE_{\hat{p}_2}(p) &= \frac{n}{4(n+2)^2} \\
 &= \frac{1}{4(n+4+4/n)}
 \end{aligned}$$

$MSE_{\hat{p}_2}(p) < MSE_{\hat{p}_1}(p)$, so we prefer $\hat{p}_2$.

2. A random sample of 500 hospital records shows that the length of stay in one of South Australia's hospitals had a (sample) mean 5.4 days and (population) standard deviation 3.1 days.
 - a. A health agency hypothesizes that the average length of stay is less than or equal to 5 days. Do the data support this hypothesis? You may use $\alpha = 0.025$. [6 marks]
 - b. How large should the sample size be if we require that $\alpha = 0.005$ and $\beta = 0.05$, assuming $\mu = 5.5$? [4 marks]

[Question total: 10]

Solutions:

a. Let μ be the true mean length of stay in the hospital. Then we are testing

$$H_0 : \mu \leq 5 \quad \text{against} \quad H_a : \mu > 5.$$

The test statistic is

$$z = \frac{\bar{y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{5.4 - 5}{\frac{3.1}{\sqrt{500}}} \approx 2.89.$$

The critical value is $z_\alpha = z_{0.025} \approx 1.96$. Hence there is sufficient evidence to reject H_0 since $z > 1.96$.

b. This is a one-sided z-test. Using the formula from lecture, we have

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{(\mu_a - \mu_0)^2} = 684.7765.$$

Hence, we need a sample of size 685 to provide the desired levels.

3. Farmer Joe is the owner of the local apple farm “*How do you like ’em...*” and they believe they have the best Pink Lady apples around. However, the local supermarket *Wooles* has decided they would prefer to start importing their apples from the next town over since they claim their apples are “Bigger and Better”. You are a staunch supporter of Farmer Joe and of apples, and you don’t think the imported apples are any different, so you’ve decide to go and test whether there is a significant difference in the weight of apples from Farmer Joe’s and *Wooles*. You collected a random sample of **44** apples from Farmer Joe’s farm, and a random sample **35** apples from *Wooles*. For the apples from Farmer Joe’s farm, you found they had a mean weight of **200.07** grams, with a variance of **2.73**. For the apples from *Wooles*, you found they had a mean weight of **199.69** grams, with a variance of **1.66**. Assuming the weight of apples from each producer is normally distributed, test the hypothesis

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs} \quad H_a : \mu_1 - \mu_2 \neq 0$$

at the $\alpha = \mathbf{0.05}$ level of significance, where μ_1 is the true mean weight of apples at Farmer Joe’s farm and μ_2 is the true mean weight of apples at *Wooles*. Complete the following sentence (give all numeric answers to 3

- For this test, we will do a 2-sample pooled or non-pooled t-test. Please describe the reason. [2 marks]
- Find the corresponding test statistic is for the test. Do you reject or retain the hypothesis in Part a? Why or why not? [7 marks]
- what is the appropriate 95% confidence interval for the difference for this test ? What is your conclusion? [3 marks]
- Find the P-value associated with this test. [5 marks]

[Question total: 17]

Solutions:

The direct solutions are

- For this test, we will do a 2-sample **pooled** t-test. Our rule of thumb to decide between pooled and not pooled is

$$\frac{\max\{s_1, s_2\}}{\min\{s_1, s_2\}} = \frac{\sqrt{2.73}}{\sqrt{1.66}} < 2.$$

- The corresponding test statistic is **1.117**, which follows a t-distribution with **77** degrees of freedom.

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)} = \frac{43(2.73) + 34(1.66)}{77} = 2.258$$

If we are doing a pooled t-test, then the appropriate test statistic is

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{200.07 - 199.69}{\sqrt{2.258(\frac{1}{44} + \frac{1}{35})}} = 1.117 \sim t_{77,0.025}.$$

Since $1.117 < t_{77,0.025}=1.99$, there is insufficient evidence to reject the the null hypothesis at the 0.05 level of significance. This means that there is insufficient evidence to suggest the true mean weight of apples from *Wooles* is any different from the true mean weight of apples from Farmer Joe’s.

- The appropriate 95% confidence interval for the difference for this test is (-0.298, 1.058).** Since, 0 is in the CI, we will not reject the null hypothesis at the 0.05 level of significance.

The appropriate CI is given by

$$\bar{X}_1 - \bar{X}_2 \pm t_{n_1+n_2-2, \alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

d. P-value associated with this test.

$$P_value = Pr(|T| > t_{obs}) = 2Pr(T > 1.117) = 0.267$$

4. Consider data $(x_1, Y_1), (x_2, Y_2), \dots, (x_{n-1}, Y_{n-1}), (x_n, Y_n)$ where $x_i \in \mathbb{R}$ are known. Let $Y_i \sim N(\mu_i, \sigma^2)$ where

$$\mu_i = \begin{cases} \beta_0 + \beta_1 x_i & \text{if } i = 1, 2, \dots, n-1, \\ \beta_0 & \text{if } i = n, \end{cases}$$

with unknown constants β_0 and β_1 . You fit a simple linear regression model to this data such that

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} \quad \text{and} \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

In the following, you may use that

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} = \sum_{i=1}^n b_i Y_i \quad \text{where} \quad b_i = \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{XX}} \right),$$

and

$$S_{XX} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \bar{x}) x_i.$$

a. Show that

$$\sum_{i=1}^n b_i = 1 \quad \text{and} \quad \sum_{i=1}^n b_i x_i = 0.$$

[6 marks]

b. Using Part (a), show that

$$E[\hat{\beta}_0] = \beta_0 - \beta_1 b_n x_n.$$

[5 marks]

c. Calculate the bias of $\hat{\beta}_0$ for β_0 given by

$$b_{\hat{\beta}_0}(\beta_0) = E[\hat{\beta}_0] - \beta_0.$$

[2 marks]

[Question total: 13]

[[Assignment total: 56]]

Solutions:

We have

$$\begin{aligned}\sum_{i=1}^n b_i &= \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{XX}} \right) \\ &= \frac{1}{n} \sum_{i=1}^n 1 - \frac{\bar{x}}{S_{XX}} \sum_{i=1}^n (x_i - \bar{x}) \\ &= \frac{1}{n} (n) - \frac{\bar{x}}{S_{XX}} (n\bar{x} - n\bar{x}) \\ &= 1,\end{aligned}$$

and

$$\begin{aligned}\sum_{i=1}^n b_i x_i &= \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{XX}} \right) x_i \\ &= \frac{1}{n} \sum_{i=1}^n x_i - \frac{\bar{x}}{S_{XX}} \sum_{i=1}^n (x_i - \bar{x}) x_i \\ &= \bar{x} - \frac{\bar{x}}{S_{XX}} S_{XX} \\ &= 0.\end{aligned}$$

Calculating the expected value:

$$\begin{aligned}\mathbb{E}[\hat{\beta}_0] &= \mathbb{E} \left[\sum_{i=1}^n b_i Y_i \right] \\ &= \sum_{i=1}^n b_i \mathbb{E}[Y_i] \\ &= \sum_{i=1}^{n-1} (b_i(\beta_0 + \beta_1 x_i)) + b_n \beta_0 \\ &= \beta_0 \sum_{i=1}^n b_i + \beta_1 \sum_{i=1}^{n-1} b_i x_i \\ &= \beta_0 + \beta_1 \sum_{i=1}^n b_i x_i - b_n \beta_1 x_n \\ &= \beta_0 - b_n \beta_1 x_n\end{aligned}$$

Hence

$$\begin{aligned}b_{\hat{\beta}_0}(\beta_0) &= \mathbb{E}[\hat{\beta}_0] - \beta_0 \\ &= \beta_0 - b_n \beta_1 x_n - \beta_0 \\ &= -b_n \beta_1 x_n.\end{aligned}$$
